



UltraRO™
Reverse Osmosis Systems
From the premier water solutions provider





UltraRO™

Ultra-Pure Water from Proven, High-Performance Systems



Nalco introduces UltraTreat Systems

UltraTreat industrial water treatment systems are components of integrated chemical, equipment, and service solutions that are a part of Nalco's philosophy of Delivering Customer Advantage™. Nalco's reliable, cost-effective, and fully integrated line of modular water filtration, ion exchange, and reverse osmosis systems can provide you with long-term competitive advantages.

UltraRO

Nalco introduces UltraRO reverse osmosis systems for reliable and efficient removal of particulates and dissolved solids from water.

Reverse osmosis is an economical and proven water treatment technology that offers benefits for almost every industry/application that uses pure water.

Industry/Application	Benefit/Operational Efficiency
Boiler/Cooling towers	Minimizes cooling tower or boiler blowdown frequency to provide energy savings.
Bottled water	Improves taste and odor. Consistent water quality. Removes harmful bacteria.
Laboratory	Final barrier for bacteria and pretreatment for deionized water.
Car wash	No spotting on vehicles during final rinse.
Metal finishing/Metalworking	Reclaims heavy metal from plating lines. Wastewater reuse. Rinse water recycling.
Pharmaceutical/Biotech	Laboratory water purification. Provides ultra-pure water for injectable drugs.
Chemical blending/Manufacturing	Consistent water quality/Maximizes efficiency of blending processes/Maximizes effectiveness of final product. Removes contaminants from recycled chemicals such as antifreeze.
Electronics manufacturing	High purity rinse (18 megaohm) leaves no residue to cause short circuits in microcircuitry.
Restaurants	Consistent, quality taste for beverages and ice. Quality, spot-free dishwashing rinse.
Food processing	Cold pasteurizes brine solutions for vegetables. Recovers starch in the potato industry.
Beverage industry	Purifies water for making beverages or juices. Filters contaminants from juices.
Automotive industry	Glass manufacturing rinse water. Rinsing parts prior to painting. Paint and solvent recovery.
Hotels/Resorts/Schools	High-purity drinking water/Removes hazardous contaminants/Reduces scale build-up in plumbing.
Dairy products	Converts whey, a cheese byproduct expensive to dispose of, into powder and water.
Laundry and washing	Removes contaminants, such as detergents and dirt, from recycled wash water.
Mining	Reclaims heavy metals from acid-water mixture to increase lime treatment efficiency.
Textile industry	Removes contaminants from dyes without using chemicals or losing any of the dye.
Pulp and Paper	Removes organic contaminants from wastewater.

UltraRO systems offer superior system quality and features

- Thin Film Composite (TFC) membranes for high performance at low pressures and temperatures
- Variable concentrate and recycle flow control valves provide performance flexibility and water savings
- Fiberglass Reinforced Plastic (FRP) housings allow easy access to membranes
- NEMA 4 and NEMA 1 electrical enclosures for safe operation in any environment
- System alarms, flow meters and conductivity meters assist in system monitoring and troubleshooting
- Compact steel frame construction accommodates small footprints

Feature-packed Nalco standard reverse osmosis systems with ranges from 576 gpd to 432,000 gpd

UltraRO operating parameters for all systems

- Operating Pressure.....150–220 psig
- Nominal Rejection.....95–99%
- Minimum Inlet Pressure.....25 psig
- Design Temperature.....77° F



UltraRO™ LI-Series

Light industrial-grade, low flow rate reverse osmosis system featuring high-rejection 8" membranes, simple operation and a microprocessor for advanced diagnostic and control functions.

Operating Parameters:

- Maximum recovery75%
- Design capacities35–165 gpm
.....50,400–237,600 gal/day

Features:

- 208, 230, 460 or 575 VAC operation
- Pretreatment lock-out and storage tank interlocks
- Automatic flush during standby
- Feed and product conductivity monitor
- Pump suction and discharge pressure switches
- Product and reject flow meters
- Pump discharge, membrane feed and reject pressure gauges
- Clean-In-Place connections
- Audible alarm and status indicators
- PVC feed and product piping
- 316L stainless steel high-pressure piping
- Stainless steel, multi-stage (horizontal) centrifugal pump
- Optional modem capability

UltraRO™ HI-Series

Heavy industrial-grade, high flow rate reverse osmosis system featuring high-rejection 8" membranes, simple operation and programmable logic controller (PLC) for advanced diagnostic and control functions.

Operating Parameters:

- Maximum recovery75%
- Design capacities37–300 gpm
.....53,280–432,000 gal/day



Features:

- 208, 230, 460 or 575 VAC operation
- Pretreatment lock-out and storage tank interlocks
- Automatic flush during standby
- Pretreatment on/off control
- Feed and product conductivity monitor
- Pump suction and discharge pressure switches
- Feed and reject flow monitor
- Pump discharge, membrane feed and reject pressure gauges
- Clean-In-Place connections
- Audible alarm and status indicators
- PVC feed and product piping
- 316L stainless steel high-pressure piping
- Stainless steel, multi-stage (horizontal) centrifugal pump

UltraRO™ H-Series

Commercial-grade, high flow rate reverse osmosis systems featuring high-rejection 4" membranes, simple operation and a microprocessor for advanced diagnostic and control functions.

Operating Parameters:

- Maximum recovery75%
- Design capacities10.5–23 gpm
.....15,120–33,120 gal/day

Features:

- 5-micron pre-filter with inlet and outlet pressure gauges
- Automatic inlet isolation valve
- NEMA 4X control enclosure
- 208, 230, 460 or 575 VAC operation
- Pretreatment lock-out and storage tank interlocks
- Automatic flush during standby
- Feed and product conductivity monitor



- Product and reject flow meters
- Pump discharge, membrane feed and reject pressure gauges
- Clean-In-Place connections
- Audible alarm and status indicators
- Stainless steel, multi-stage (vertical) centrifugal pump
- Optional modem capability for remote monitoring and control



UltraRO™ L-Series

Commercial-grade, low flow rate reverse osmosis systems featuring low-energy 2 1/2" membranes and simple operation. A vertical configuration is employed for a small footprint.

Operating Parameters:

- Maximum recovery50%
- Design capacities0.4–1.6 gpm
.....576–2,304 gal/day

Features:

- 5-micron pre-filter with inlet and outlet pressure gauges
 - Automatic inlet isolation valve
 - NEMA 4X motor starter with thermal protection
 - 120 VAC operation
 - Product, reject and recycle flow meters
 - Pump discharge and reject pressure gauges
 - Clean-In-Place connections
 - Optional product conductivity monitor
- (Shown with optional controller)



UltraRO™ V-Series

Commercial-grade, medium flow rate reverse osmosis systems featuring low-energy 4" membranes, simple operation and a microprocessor for advanced diagnostic and control functions. A vertical configuration is employed for a small footprint.

Operating Parameters:

- Maximum recovery50%–75%
- Design capacities1.4–10 gpm
.....2,016–14,400 gal/day

Features:

- 5-micron pre-filter with inlet and outlet pressure gauges
- Automatic inlet isolation valve
- NEMA 4X control enclosure
- 208, 230, 460 or 575 VAC operation
- Pretreatment lock-out and storage tank interlocks
- Automatic flush during standby
- Feed and product conductivity monitor
- Product, reject and recycle flow meters
- Pump discharge, membrane feed and reject pressure gauges
- Clean-In-Place connections
- Audible alarm and status indicators
- Stainless steel, multi-stage (vertical) centrifugal pump
- Optional modem capability for remote monitoring and control

Nalco Company

Nalco Center
Naperville, Illinois
60563-1198 USA
T 630-305-1000
www.nalco.com



Theory of Operation

UltraRO reverse osmosis systems are designed to reject dissolved mineral salts, organic molecules and certain other impurities from water by forcing water to pass through a semi-permeable membrane. Solids-free water exiting the UltraRO system is called the permeate, and can range from 50-75% of the influent water flow. Water that is loaded with solids exits the UltraRO system as waste water, called the concentrate.

The permeate flow is mainly a function

of water temperature and system pressure, while the water quality of the permeate is dependent on the concentration of the dissolved solids entering the system.

Pretreatment of water prior to the UltraRO system is normally required. High water hardness, chlorine level, turbidity and iron levels all have negative impacts on membrane performance and useful life.

Nalco uses Thin Film Membranes in all its UltraRO systems. These membranes have a high mineral rejection rate, with good performance at wide pH and temperature ranges.